Threshold Deal Logic — Farmer Deal Creation

HarvestHub · October 10, 2026 · Daejon

Purpose and locked decisions

A farmer sets a regular price, a lower deal price, a household goal and a deadline; the deal price unlocks only if enough households commit before the deadline. Everything below builds on decisions already made, and any number marked *proposed* still needs your sign-off.

- Commitments count by household; each Pod member household counts once
- Subscribers get early access before the public window opens
- No cap on how many deals a farmer runs at once
- Produce only at launch
- If charge failures drop an unlocked deal below goal, it is honored by default
- A regular-price fallback offer goes to subscribers only when a deal misses
- Pods see their combined contribution on the progress bar
- 10% commission applies to the line-item subtotal, as with regular orders

Deal fields and validation

A farmer fills in ten fields; most are prefilled from the listing or smart defaults, so most deals take under two minutes. Hard rules block publishing; soft rules show a warning but let the farmer proceed.

Field	What the farmer sets	Hard rule	Soft rule / default	Status
Product	A produce listing from active inventory	Produce category only	Prefilled from the listing they start from	Locked
Unit	Bundle size, e.g. 5 lb box	Must match a listing unit	Default: the listing's main unit	Proposed
Regular price	Per-unit price	Pulled from the listing; frozen when the deal goes live	—	Proposed
Deal price	Discounted per-unit price	10–40% below regular price	Suggest 15–25%	Decided
Household goal	Households needed to unlock	Minimum 5; must be ≤ capacity	Warn if > 1.5× the farmer's recent weekly households	Decided
Capacity	Most households the farmer can supply	≥ goal; units reserved from inventory	Default: goal × 1.5	Proposed
Max units per household	Units one household may claim	1–3	Default: 1	Proposed
Commitment window	Public start and deadline	3–14 days; deadline at 8 pm ET	Presets: 3, 5, 7 days	Decided
Fulfillment date	Pickup or delivery day	1–10 days after deadline	Default: next delivery day after deadline	Proposed

Field	What the farmer sets	Hard rule	Soft rule / default	Status
Subscriber fallback	Offer regular price to subscribers if missed	Yes / no	Default: on	Locked (subscribers only)

Subscriber early access opens 24 hours before the public start, so the farmer sets one start date and the system derives both windows.

Pricing guardrails and payout math

The creation form shows the farmer's net payout at goal and at capacity before they publish, so the discount decision is made in dollars, not percentages. Net payout per unit is the deal price minus the 10% commission.

Net per unit = Deal price × 0.90
Net at goal = G × u × Deal price × 0.90

Where G is the household goal and u is units per household.

Example: 5 lb tomato box	Regular sale	Deal at goal	Deal at capacity
Price per box	$30.00	$24.00 (20% off)	$24.00
Households	—	25	38
Farmer net per box	$27.00	$21.60	$21.60
Farmer net total	—	$540.00	$820.80

Guardrails:

- **Discount band 10–40%**: under 10% gives households little reason to rally; over 40% risks undercutting the farmer's own regular listings
- **Goal ≤ capacity, capacity ≤ reserved inventory**: units are reserved when the deal goes live, so a regular-order shopper can never buy the last box a deal needs
- **Price freeze**: once live, regular price, deal price and goal cannot change, so every committed household sees the same terms
- **Why this price**: the consumer card shows regular price, deal price and the household savings, consistent with HarvestHub's transparency principle

Counting commitments

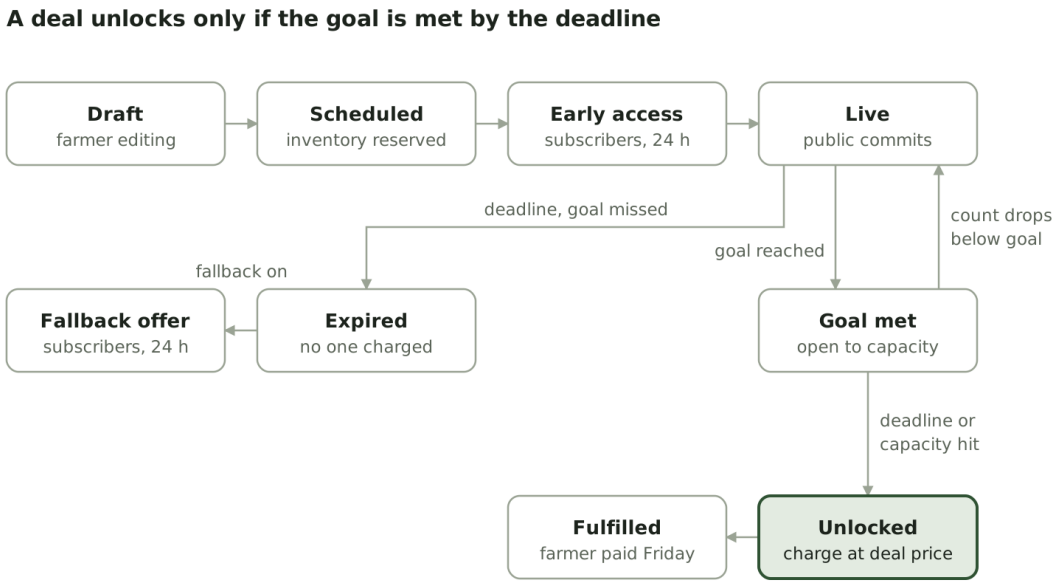
The goal counts households, not units: a household claiming 3 boxes moves the bar by 1 but draws 3 units from capacity.

- **Household identity**: one count per delivery address; a second account at the same address with the same payment method is treated as the same household (proposed dedupe rule)
- **Pods**: each member household commits and counts individually; the bar shows a highlighted segment for the Pod's combined count
- **Subscribers**: commits made during early access count toward the goal like any other
- **Payment at commit**: the payment method is saved and verified, but nobody is charged until the deal unlocks; card authorization holds usually lapse within about a week, so holds are not relied on for 14-day windows
- **Withdrawing**: a household can withdraw freely until 12 hours before the deadline, then the commitment is locked; if a withdrawal drops an unlocked count below goal, the deal relocks until the count recovers

- **Capacity check**: a commit is refused once committed units would exceed capacity; the household can join a waitlist that fills any withdrawals

Deal lifecycle

Every deal moves through nine states, and only the deadline or a full capacity can move it to Unlocked or Expired.



Deal lifecycle · 9 states

A farmer can cancel from Draft through Goal met; after unlocking, cancellation needs a stated reason (see the next section).

Deadline evaluation, charging and failures

At the deadline the system freezes commits, compares the household count to the goal, and either charges everyone at the deal price or charges no one.

1. **Freeze** at the deadline (8 pm ET): no new commits or withdrawals.
2. **Count** committed households after withdrawals and dedupe.
3. **Goal met** → status Unlocked; charge each household deal price × units, plus delivery and tax.
    a. Failed charges get one automatic retry and a 24-hour window for the household to update payment (proposed).
    b. If failures leave the paid count below goal, the deal is still honored at the deal price for everyone who paid; the farmer sells fewer boxes and HarvestHub does not top up the difference.
    c. The farmer is paid only for successful charges, in the Saturday–Friday period of the order's completion date.
4. **Goal missed** → status Expired; no one is charged and reserved units return to inventory.
    a. If the farmer enabled fallback, committed subscribers get a 24-hour offer to buy at regular price, up to capacity (proposed).
    b. Non-subscribers get a "didn't make it this time" notice and a follow on the farmer.

5. **Capacity hit early** → the deal closes before the deadline and goes straight to Unlocked.

Farmer controls once live (proposed):

- **One extension** of up to 48 hours, only if the count is at least 70% of goal and the deadline is more than 24 hours away
- **Capacity increase** at any time, if inventory allows; never a decrease below committed units
- **Cancel before goal met** without penalty; **cancel after goal met** only with a stated reason such as crop loss; households are refunded in full and the cancellation is logged against the farmer's trust status

Farmer guidance in the creation flow

The flow fits on one field-ledger screen with three steps — what, how many, when — and a live payout panel beside it, in keeping with the farmer brevity principle.

1. **What**: pick the listing; regular price and unit fill in; slide to a deal price and see the percent off and net per unit update.
2. **How many**: a suggested goal from recent demand signals, e.g. "You averaged 18 households a week; a goal of 20 is reachable"; capacity defaults to goal × 1.5.
3. **When**: choose a 3, 5 or 7-day preset; fulfillment date fills to the next delivery day; fallback toggle on by default.

The payout panel shows net at goal, net at capacity and the regular-sale comparison in Roboto Mono. Warnings appear inline in Beet, never as blocking modals, except for the hard rules in the fields table.

Reference pseudocode

Three functions carry the logic: validate on publish, accept or refuse a commit, and evaluate at the deadline. Constants are the values above.

```
const RULES = {
  minDiscount: 0.10, maxDiscount: 0.40,
  minGoal: 5, maxUnitsPerHousehold: 3,
  minWindowDays: 3, maxWindowDays: 14,
  earlyAccessHours: 24, withdrawLockHours: 12,
  commission: 0.10,
};

function validateDeal(d: DealDraft): { errors: string[]; warnings: string[] } {
  const errors = [], warnings = [];
  const discount = 1 - d.dealPrice / d.regularPrice;

  if (d.listing.category !== 'produce') errors.push('Produce only at launch');
  if (discount < RULES.minDiscount || discount > RULES.maxDiscount)
    errors.push('Deal price must be 10–40% below regular');
  if (d.goal < RULES.minGoal) errors.push('Goal must be at least 5 households');
  if (d.capacity < d.goal) errors.push('Capacity must be at least the goal');
  if (d.capacity * d.maxUnits > d.listing.availableUnits)
    errors.push('Not enough inventory to reserve for capacity');

  const days = daysBetween(d.publicStart, d.deadline);
  if (days < RULES.minWindowDays || days > RULES.maxWindowDays)
    errors.push('Window must be 3–14 days');

  if (d.goal > 1.5 * d.farmer.avgWeeklyHouseholds)
    warnings.push('Goal is well above your usual weekly households');
  return { errors, warnings };
}
```

```
function canCommit(deal: Deal, h: Household, units: number, now: Date): boolean {
  if (!['early_access', 'live', 'goal_met'].includes(deal.status)) return false;
  if (deal.status === 'early_access' && !h.isSubscriber) return false;
  if (deal.hasHousehold(h.addressId)) return false;          // one count per household
  if (units < 1 || units > deal.maxUnits) return false;
  return deal.committedUnits + units <= deal.capacity * deal.maxUnits;
}

function evaluateAtDeadline(deal: Deal): void {
  deal.freeze();
  const households = deal.activeCommitments().length;

  if (households >= deal.goal) {
    deal.status = 'unlocked';
    for (const c of deal.activeCommitments()) chargeWithRetry(c, deal.dealPrice);
    // honored even if failures drop paid count below goal
  } else {
    deal.status = 'expired';
    deal.releaseInventory();
    if (deal.subscriberFallback) offerRegularPrice(deal.subscriberCommitments(), 24);
    notify(deal.nonSubscriberCommitments(), 'missed');
  }
}
```

Edge-case decisions

All seven edge-case questions were settled on October 10, 2026, and the sections above reflect them.

- ☑️ Honored below goal: the farmer sells fewer boxes at the deal price; HarvestHub does not top up the difference
- ☑️ Numbers confirmed: 10–40% discount band, minimum goal of 5, 3–14 day window, 24-hour early access, 12-hour withdrawal lock
- ☑️ A farmer can run two live deals on one listing; inventory reservation counts both
- ☑️ A household that leaves its Pod mid-deal keeps its commitment as an individual household
- ☑️ SNAP/EBT households can commit to deals
- ☑️ Crop loss after unlock: households get a full refund, and the cancellation is logged against the farmer’s trust status
- ☑️ The deadline is fixed at 8 pm ET for every deal